

1 Topological Spaces

Definition 1.0.1. Let X be any set. A topology T on X is a collection of subsets of X , which satisfies the following axioms.

- $\emptyset$ and X are both in T
- Closed under arbitrary unions
- Closed under finite intersections

The sets in T are called open sets.

A set X equipped w/ a topology is called a topological space.

Example 1.0.2.

- X is any set, $T = P(X)$. (discrete topology)
- X is any set, $T = \{\emptyset, X\}$. (indiscrete topology)
- $X = \{a, b, c\}$, $T = \{\emptyset, X, \{a, b\}\}$
- $X = \mathbb{R}$, $T = \{\emptyset, X\} \cup \{U \subset \mathbb{R} \mid \text{for all } x \in U, \exists \varepsilon > 0 \text{ s.t. } (x - \varepsilon, x + \varepsilon) \subset U\}$

Proposition 1.0.3. *This is a topology.*

Proof.

- $\emptyset, \mathbb{R}$ are both open
- Suppose $\{U_j\}_{j \in J}$ is a collection of open sets in $\mathbb{R}$
Let $x \in \bigcup_{j \in J} U_j$. Then $x \in U_j$ for some $j \in J$.
 U_j is open thus there is an open interval in U_j containing x . So it is in the arbitrary union
- Let U, V be open subsets of $\mathbb{R}$
Let $x \in U \cap V$. Because U, V open there are intervals $(a, b) \subset U$, $(c, d) \subset V$ s.t. $x \in (a, b) \cap (c, d)$
 $(a, b) \cap (c, d) = (\max(a, c), \min(b, d)) \subset U \cap V$. Thus, x is in this interval thus $U \cap V$ is open.

□

- X is any set, let T be the collection of subsets $U \subseteq X$ s.t. either $U = \emptyset$, or $X \setminus U$ is finite.

Proposition 1.0.4. *This is a topology. (cofinite topology)*

Proof.

- The empty set is open, and $X \setminus X = \emptyset$ is finite, so $X \in T$
- Consider sets $\{U_j\}_{j \in J}$ s.t. $U_j \in T$
 First: consider the case for some j' , $X \setminus U_{j'}$ is finite.
 $X \setminus \bigcup_{j \in J} U_j = \bigcap_{j \in J} X \setminus U_j \subset X \setminus U_{j'}$. So the arbitrary union is finite.
 Second: If the first case does not occur, then $U_j = \emptyset$ for all $j \in J$, so the union is the empty set.
- Let $U, V \in T$. If either is the empty set then $U \cap V = \emptyset \in T$, so assume that both are not empty. Then consider $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$, so it is a union of two finite sets which is finite thus $U \cap V \in T$

□

Definition 1.0.5. If X is a set with topology T , then the complement of an open set in X is called a closed set.

Example 1.0.6. If T is cofinite topology, a set in $A \subseteq X$ is closed $\iff$ either finite or $A = X$

IMPORTANT: Some sets are neither open nor closed. Some sets are both closed and open.